



**INTRODUCTORY PHYSICS LABORATORY
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NAME :
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EXPERIMENT: PHYSICAL PENDULUM

Objectives:

1. To determining Earth's acceleration due to gravity using a physical pendulum.

EXPERIMENT SUMMARY

INTRODUCTION

Simple Harmonic Motion (SHM) is a form of periodic motion commonly observed in various physical phenomena. This motion occurs when an object is subjected to a restoring force that always points toward the equilibrium point and whose magnitude is proportional to the object's displacement from its equilibrium position. As a result of this restoring force, the object moves back and forth periodically around the equilibrium point. Prinsip GHS menjadi dasar dalam mempelajari berbagai sistem osilasi, seperti sistem pegas, bandul matematis, maupun bandul fisis. The SHM principle serves as the foundation for studying various oscillatory systems, such as spring systems, mathematical pendulums, and physical pendulums [1].

One system that approximates simple harmonic motion is a physical pendulum, which is a rigid body that swings about a horizontal axis of rotation. Unlike a mathematical pendulum, which assumes that the object's entire mass is concentrated at a single point, in a physical pendulum the mass is distributed throughout the object, so the mass distribution affects the characteristics of its swing. When a pendulum is displaced from its equilibrium position and then released, the force of gravity generates a torque that acts as a restoring force, causing the object to oscillate periodically. In addition to being influenced by Earth's acceleration due to gravity, the period of a physical pendulum's swing also depends on its moment of inertia and the position of its center of mass relative to the suspension point [2]. In a physical pendulum, the weight mg acting at the center of mass produces a torque about the suspension point equal to:

$$\tau = -mga \sin \theta \quad (1)$$

for small deviations, the approximation $\sin \theta \approx \theta$ applies, so the torque equation becomes:

$$\tau = -mga\theta \quad (2)$$

according to Newton's second law of rotational motion:

$$\tau = I\alpha \quad (3)$$

where (I) is the moment of inertia of the object about the suspension point. Thus, the equation of motion for a physical pendulum is written as follows:

$$I \frac{d^2\theta}{dt^2} + mga\theta = 0 \quad (4)$$

which is the differential equation of simple harmonic motion for a rotating system. Solving this equation yields the period of the physical pendulum's swing:

$$T = 2\pi \sqrt{\frac{I}{mga}} \quad (5)$$

where (T) is the period of oscillation (s), (I) is the moment of inertia about the suspension point ($kg \cdot m^2$), (m) is the mass of the object (kg), (a) is the distance from the center of mass to the suspension point (m), and (g) is the acceleration due to gravity on Earth (m/s^2). This equation shows that the period of oscillation is influenced by the mass distribution of the object, the position of the center of mass, and the acceleration due to gravity on Earth [3]. Earth's gravitational acceleration is determined by measuring the period of oscillation at two different suspension points. The value of gravitational acceleration is obtained using the relationship :

$$\frac{T_1^2 + T_2^2}{8(a_1 + a_2)} + \frac{T_1^2 - T_2^2}{8(a_1 - a_2)} = \frac{\pi^2}{g} \quad (6)$$

where, (T_1) and (T_2) are the oscillation periods at suspension points A and B, respectively, while (a_1) and (a_2) are the distances from the center of mass to each suspension point. This equation allows the value of Earth's acceleration due to gravity to be calculated based on measurements of the oscillation periods and the position of the center of mass of the physical pendulum system. The physical pendulum experiment was conducted to determine the value of Earth's gravitational acceleration by measuring the period of oscillation of a rigid body and to compare the results obtained with the standard value of Earth's gravitational acceleration [4].

METHODOLOGY

A. Tools and Materials

Listed below are the tools and materials used in the experiment:

Table 1. Tools and materials for the physical pandulum experiment.

Apparatus	Quantity	Function
Physical pendulums set	1 set	Used as a pendulum system to observe oscillatory motion and determine the period of oscillation and the moment of inertia of the physical pendulum.
Measuring rod	1 set	Used to suspend the physical pendulum and determine the angle of inclination.
Ruler	1 piece	Used to measure the length and the distance from the suspension point to the center of mass.
Stopwatch	1 piece	Used to measure the time it takes for the pendulum to complete a certain number of oscillations.

B. Procedure

The steps for this experiment are as follows.:

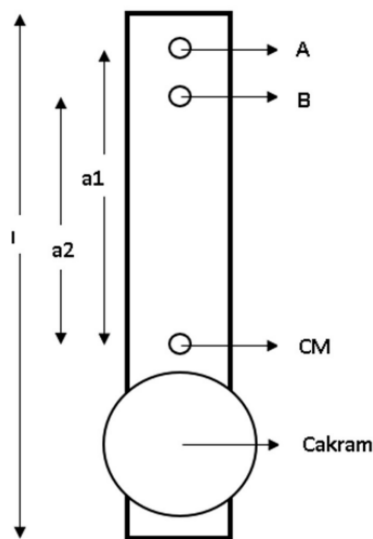


Figure 1. Illustration of a physical pendulum

1. Place the disc in a specific position and find the centre of mass (CM) for that position. Bear in mind that the location of the CM always changes depending on the position of the disc.
2. To find the position of the centre of mass (CM), use your fingers to estimate the ratio and then mark the CM point.
3. Measure a_1 , which is the distance between point A and the center of mass (CM), using a ruler.
4. Hang the pendulum (a rod with a disc weight attached) from point A as shown in Figure 1 and measure a_1 .
5. Swing the rod with a small amplitude and record the time taken for 4 oscillations.
6. Select another point, B, as the suspension point and measure a_2 . Repeat steps 1–4.
7. Repeat the experiment for different angles of deviation in accordance with the provided practical worksheet.

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement results obtained in the laboratory:

Table 2. Results of physical pendulum measurements at a 10° angle variation

Repetition	A period of 4 oscillations			
	$a_1(m)$	$t_1(s)$	$a_2(m)$	$t_2(s)$
1				
2				
3				
4				
5				
Average				

Table 3. Results of physical pendulum measurements at a 15° angle variation

Repetition	A period of 4 oscillations			
	$a_1(m)$	$t_1(s)$	$a_2(m)$	$t_2(s)$
1				
2				
3				
4				
5				
Average				

Approved by:

Laboratory Assistant

B. Calculation Analysis

(Calculate the acceleration due to gravity using Equation 6)
The following is a sample calculation based on experimental data.
Sample Calculation:

Using the same method, the calculation above yields the following value for the acceleration due to gravity:

Table 3. Results of the physical pendulum calculations for a 10° angle variation

Repetition	A period of 4 oscillations		$g \text{ (m/s}^2\text{)}$
	$T_1\text{(s)}$	$T_2\text{(s)}$	
1			
2			
3			
4			
5			
Average			

Table 4. Results of the physical pendulum calculations for a 15° angle variation

Repetition	A period of 4 oscillations		$g \text{ (m/s}^2\text{)}$
	$T_1\text{(s)}$	$T_2\text{(s)}$	
1			
2			
3			
4			
5			
Average			

C. Discussions

CONCLUSIONS

From this experiment, it can be concluded that:

- 1.

REFERENCES

- [1] D. Halliday, R. Resnick, and J. Walker, Fundamentals of Physics Extended, 9th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [2] D. C. Giancoli, Physics: Principles with Applications, 7th ed. Upper Saddle River, NJ, USA: Pearson Education, 2014.
- [3] R. A. Serway and J. W. Jewett, Physics for Scientists and Engineers with Modern Physics, 9th ed. Mason, OH, USA: Brooks/Cole, Cengage Learning, 2012.
- [4] M. Abdullah, Fisika Dasar I, Edisi Revisi. Bandung, Indonesia: Penerbit ITB, 2016.

PRELIMINARY ASSIGNMENT

1. Prove equation (6) and derive the equation of gravity!
2. Explain what is meant by the centre of mass and why its position affects the period of a physical pendulum's swing.
3. Why is the swing time measured over several oscillations, and then divided by the number of oscillations to obtain the period?
4. Explain the factors that influence the period of a physical pendulum's swing.

